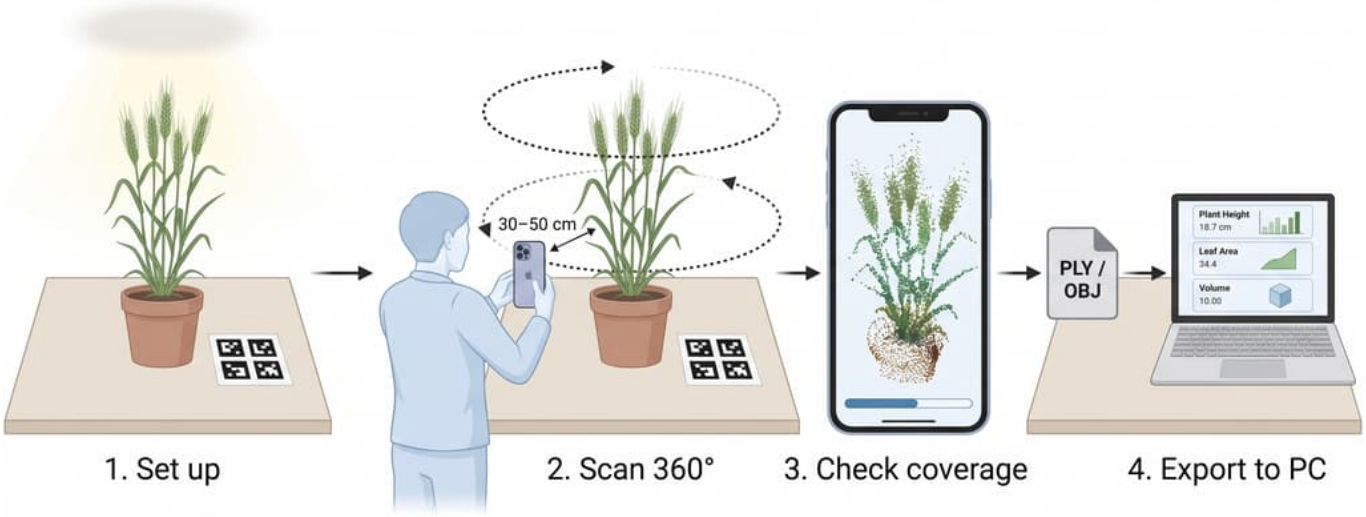
SOP-1 iPhone / iPad Pro with LiDAR

Metric 3D scans of single plants and trays in a few minutes

SOP-1 · PhenoForge capture SOP · version 1.0 (October 2026)



What you need

Checklist

- iPhone 12 Pro or later Pro / Pro Max, or iPad Pro (2020+) with LiDAR
- A LiDAR scanning app that exports PLY or OBJ (e.g. 3d Scanner App, Polycam, Scaniverse, Record3D)
- ArUco card (50 mm markers) and the plant label
- Matte surface, diffuse light
- USB cable, AirDrop to a Mac, or cloud folder to move files to the PC

Capture

1. **Set up**
   Put the pot on a matte surface with the ArUco card flat beside it. Leave 50 cm of free space around the plant to walk.

2. **Choose the mode**
   In the app use the **LiDAR / point cloud** mode with the highest resolution and confidence; for thin wheat leaves prefer *high resolution, low max depth (≈1 m)*.

3. **Scan 360°**
   Hold the phone 30–50 cm from the plant. Walk one slow circle at pot height, then a second circle from above (about 45°). Keep the plant in the centre of the screen; a full circle should take 40–60 s.

4. **Check coverage**
   Look at the preview: no grey holes on the leaves, the markers visible and sharp. Re-scan rather than patch.

5. **Export**
   Export **PLY (point cloud)** or **OBJ / USDZ (mesh)** with colour, in metres. Name the file with the plant ID and date, e.g. *W123_2026-10-19.ply*, and copy it to the PC.

In PhenoForge

1. **Import**
   Tools > Import from a capture device, device *iPhone / iPad with LiDAR*, choose the file, **Import**. Or open it directly in the 3D analysis tab.

2. **Keep the LiDAR scale, but verify it**
   Scale source *Already metric*. Click **Markers image**: each marker should read within ±2 % of the printed size. If not, switch the scale source to **ArUco markers**.

3. **Segment**
   Section 3: *single* for one plant, *watershed* for a tray of touching plants (enter the number of plants), or *grid* with rows, columns and pot pitch for a bench of equally spaced pots.

4. **Run and export**
   Press **Run analysis**. Results appear in the Scan / Per plant / Leaves tables; **Save results** writes JSON and CSV. Colour the view by Height, Greenness or Plants to check the segmentation.

Quality checks

Check	Pass	If it fails
Marker size after scaling	within ±2 %	Use ArUco scaling; check that the card is flat
Plant height vs ruler	within 1 cm	Fit ground plane on soil, not on leaves (crop the scan)
Leaves look solid, not double	single surface	Scan slower; avoid wind; re-scan
Points on the plant	> 50 000	Move closer; use a higher resolution mode

Limits of phone LiDAR

- The LiDAR depth grid is coarse (≈ 256×192 per frame); apps fuse it with the RGB camera. Thin awns, fine tillers and leaves narrower than ~3 mm are often lost or thickened.
- Wheat heads and thin leaves are better captured by photogrammetry (SOP-2 or SOP-3) with ArUco scaling.